FOUNDATION COURSE

START YOUR
PREPARATION
FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 17

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Recap *of previous lecture*

MAXIMA - MINIMA



Topics to be covered

"INDEFINITE INTEGRATION"



Derivative

$\frac{d}{dx}(x^3) = 3x^2 \Rightarrow \int (3x^2)\,dx = x^3$

Antiderivative

$\frac{d}{dx}(x^3+5) = 3x^2 \Rightarrow \int (3x^2)\,dx = x^3+5$

$\frac{d}{dx}(x^3-7) = 3x^2 \Rightarrow \int (3x^2)\,dx = x^3-7$

$\frac{d}{dx}(x^3+4) = 3x^2$

$\frac{d}{dx}(x^3+c) = 3x^2 \Rightarrow \boxed{\int (3x^2)\,dx = x^3+C}$

ie Indefinite Int is the Collection of all Antiderivatives (Primitives) of func^n — Defn

ie In General if

$\frac{d}{dx}(f(x)+C) = \phi(x)$

$\Rightarrow \boxed{\int \phi(x)\,dx = f(x)+C}$

Here $\phi(x)$ = Derivative of $f(x)+C$

& $f(x)+C$ = Antiderivative of $\phi(x)$

Some Standard Results :-

(1) (i) $\int x^a dx = \frac{x^{a+1}}{a+1} + C$; $a \neq -1$

(2) $\int a^x dx = \frac{a^x}{\log_e a} + C$

(3) $\int e^x dx = e^x + C$

(4) $\int \sin x dx = -\cos x + C$

(5) $\int \cos x dx = \sin x + C$

(6) $\int \tan x dx = \log \sec x + C$

(7) $\int \cot x dx = \log \sin x + C$

(8) $\int \operatorname{cosec} x dx = \log(\operatorname{cosec} x - \cot x) + C$

(9) $\int \sec x dx = \log(\sec x + \tan x) + C$

(10) $\int \sec^2 x dx = \tan x + C$

(11) $\int \operatorname{cosec}^2 x dx = -\cot x + C$

(12) $\int \sec x \tan x dx = \sec x + C$

(13) $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$

(1) (ii) $\int \left(\frac{1}{x}\right) dx = \log_e x + C$

(14) $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+c$

(15) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left(\frac{x-a}{x+a}\right)+c$

(16) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left(\frac{a+x}{a-x}\right)+c$

(17) $\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left(x+\sqrt{x^2+a^2}\right)+c$

(18) $\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left(x+\sqrt{x^2-a^2}\right)+c$

(19) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)+c$

(20) $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left(x+\sqrt{x^2+a^2}\right)+c$

(21) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left(x+\sqrt{x^2-a^2}\right)+c$

(22) $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+c$

(23) $\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)+c$

(24) $\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)+c$

(23) Integration by parts $\Rightarrow \int \underset{\text{I}}{u} \cdot \underset{\text{II}}{v}\,dx = u\int v\,dx - \int \left\{\frac{d}{dx}(u)\cdot\int v\,dx\right\}dx + c$

Log func^n — Trig func^n

I L A T E

Inverse Trig func^n — Algebraic func^n — Exp. func^n

The letter which is coming 1st in the word ILATE should be assumed as 1st func^n u

(*) $\int \frac{f'(x)}{f(x)}\,dx = \log f(x) + C$

Put $f(x) = t \Rightarrow f'(x)\,dx = dt$

$\therefore I = \int \frac{dt}{t} = \log t = \log f(x) + C$

eg $\int \cot x\,dx = ? = \int \frac{\cos x}{\sin x}\,dx = \log(\sin x) + C$

eg $\int \tan x\,dx = ? = -\int \frac{-\sin x}{\cos x}\,dx = -\log\cos x + C$

$= \log(\cos x)^{-1} = \log(\sec x) + C$

(*) $\int e^x \{f(x) + f'(x)\}\,dx = e^x f(x) + C$

eg $\int e^x \{\sin x + \cos x\}\,dx = ? = e^x \sin x + C$

(*) $\int f^n(x)\, f'(x)\,dx = \frac{f^{n+1}(x)}{n+1} + C$

eg $\int \sec^2 x\,(\sec x \tan x)\,dx = \frac{\sec^3 x}{3} + C$

eg $\int (\log x)^3 \left(\frac{1}{x}\right) dx = \frac{(\log_e x)^4}{4} + C$

Methods of Solving Questions → (12th Class) → There are Four Methods →

1. M-I → By using Standard Results
2. M-II → By using Substitution Method (Imp)
3. M-III → By using Integration By Parts
4. M-IV → By using partial fractions

Quest Based on M.I.

$$\int x^a dx = \frac{x^{a+1}}{a+1} + C, \quad a \neq -1$$

i.e. $\int k\,dx = kx + C$, & $\int \frac{1}{\sqrt{x}}dx = 2\sqrt{x} + C$

Eg $\int 1\,dx = \int x^0 dx = \frac{x^{0+1}}{0+1} = \frac{x^1}{1} = x + C$

Eg $\int x^1 dx = \frac{x^2}{2} + C$

Eg $\int x^2 dx = \frac{x^{2+1}}{2+1} = \frac{x^3}{3} + C$

Eg $\int \sqrt{x}\,dx = \int x^{\frac{1}{2}} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C$

Eg $\int \frac{dx}{\sqrt{x}} = \int x^{-\frac{1}{2}} dx = \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = 2\sqrt{x}$

(P8) $\int \left(x^a + a^x + a^a + \frac{x}{a} + \frac{a}{x}\right) dx = ?$

$= \frac{x^{a+1}}{a+1} + \frac{a^x}{\log_e a} + a^a(x) + \frac{1}{a}\left(\frac{x^2}{2}\right) + a\left(\log_e x\right) + C$

Eg $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 dx = ?$

$= \int \left(x + \frac{1}{x} + 2\right) dx = \frac{x^2}{2} + \log x + 2x + C$

Q. $\int \left(\frac{e^{5\log x} - e^{4\log x}}{e^{3\log x} - e^{2\log x}} \right) dx = ?$

$$= \int \left[\frac{e^{\log_e(x^5)} - e^{\log_e(x^4)}}{e^{\log_e(x^3)} - e^{\log_e(x^2)}} \right] dx$$

$$= \int \left(\frac{x^5 - x^4}{x^3 - x^2} \right) dx = \int \frac{x^4}{x^3} \frac{(x-1)}{(x-1)} dx$$

$$= \int x\, dx = \frac{x^2}{2} + c$$

Q. $\int \frac{1}{\sin^2 x \cos^2 x} dx = ?$

$$= \int \left(\frac{\sin^2 x + \cos^2 x}{\sin^2 x \cos^2 x} \right) dx$$

$$= \int (\sec^2 x + \operatorname{cosec}^2 x) dx$$

$$= \tan x - \cot x + c$$

Q. $I = \int \frac{x^4}{x^2+1} dx = ?$

Sol. $I = \int \left(\frac{x^4 - 1 + 1}{x^2+1} \right) dx = \int \frac{(x^2-1)(x^2+1)}{(x^2+1)} dx + \int \frac{1}{x^2+1} dx = \int (x^2-1) dx + \int \frac{1}{x^2+1} dx = \frac{x^3}{3} - x + \tan^{-1} x + c$

Questions Based on (M-II)

Q. $\int x^2 \cdot \sin(x^3)\,dx = ?$

Put $x^3 = t$

$\frac{d}{dx}(x^3) = \frac{d}{dx}(t)$

$3x^2 = \frac{dt}{dx} \Rightarrow x^2\,dx = \frac{1}{3}\,dt$

Note: Put $x^3 = t \Rightarrow 3x^2\,dx = dt \Rightarrow x^2\,dx = \frac{dt}{3}$

$\therefore I = \int \sin(t) \cdot \frac{dt}{3} = \frac{1}{3}(-\cos t) = -\frac{1}{3}\cos(x^3) + C$

Q. $I = \int \sec^3 x \tan x\,dx = ?$

$= \int \sec^2 x \cdot (\sec x \tan x)\,dx$

Put $\sec x = t \Rightarrow \sec x \tan x\,dx = dt$

$\therefore = \int t^2\,(dt) = \frac{t^3}{3} = \frac{\sec^3 x}{3} + C$

Q. $I = \int \frac{(\log x)^3}{x}\,dx = ?$

Put $\log x = t \Rightarrow \frac{1}{x}\,dx = dt$

$\therefore I = \int t^3\,dt = \frac{t^4}{4} = \frac{(\log x)^4}{4} + C$

Q. $I = \int \sin(\log x)\, dx = ?$

Sol: Put $\log_e x = t \Rightarrow x = e^t$

$dx = e^t \cdot dt$

$I = \int \sin(t) \cdot e^t\, dt$

$= \int e^{1t} \cdot \sin 1t\, dt$

$\int e^{ax} \sin bx\, dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$

$I = \frac{e^t}{1^2+1^2}(1\cdot\sin t - 1\cdot\cos t) = \frac{e^t}{2}(\sin t - \cos t)$

Q. $I = \int \frac{e^x}{\sqrt{4-e^{2x}}}\, dx = ?$

Put $e^x = t \Rightarrow e^x \cdot dx = dt$

$I = \int \frac{dt}{\sqrt{4-t^2}} = \int \frac{1}{\sqrt{(2)^2-t^2}}\, dt$

w.k.t $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

$I = \sin^{-1}\left(\frac{t}{2}\right) + C = \sin^{-1}\left(\frac{e^x}{2}\right) + C$

Q. $\int \sec x\,dx = ?$

$$I = \int \sec x \left(\frac{\sec x + \tan x}{\sec x + \tan x}\right) dx$$

$$= \int \frac{(\sec^2 x + \sec x \tan x)}{\tan x + \sec x}\, dx$$

$$= \int \frac{dt}{t} = \log t = \log(\sec x + \tan x) + C$$

AW $\int \operatorname{cosec} x\,dx = ? = \log(\operatorname{cosec} x - \cot x) + C$

Q. $I = \int \frac{dx}{1+e^{-x}} = ?$

$$I = \int \left(\frac{e^x + 0}{e^x + 1}\right) dx = \log(e^x + 1) + C$$

Q. $\int \frac{x e^x}{(x+1)^2}\, dx = ? = \int e^x \left\{\frac{(x+1)-1}{(x+1)^2}\right\} dx$

$$= \int e^x \left\{\frac{1}{(x+1)} - \frac{1}{(x+1)^2}\right\} dx$$

$$= \int e^x \left\{\underbrace{\left(\frac{1}{(x+1)}\right)}_{f(x)} + \underbrace{\left(\frac{-1}{(x+1)^2}\right)}_{f'(x)}\right\} dx$$

$$= e^x \left(\frac{1}{x+1}\right) + C$$

Q. $\int e^x \left\{ \frac{1+\sin x \cos x}{\cos^2 x} \right\} dx = ? = \int e^x \{ \sec^2 x + \tan x \} dx = \int e^x \{ \tan x + \sec^2 x \} dx$

$= e^x \tan x + C$ //

Quest Based on M-II

(*) $\int uv\,dx = u \int v\,dx - \int \left\{ \frac{du}{dx} \int v\,dx \right\} dx + C$

$\int x^2 e^x dx = ? = x^2 \int e^x dx - \int \{ (2x)(e^x) dx + C$

(x^2 : u, e^x : v)

$= x^2 (e^x) - 2 \int x e^x dx + C$

(x : u, e^x : v)

$= x^2 (e^x) - 2 \left[x(e^x) - \int (1)(e^x) dx \right] + C$

$= x^2 e^x - 2xe^x + 2e^x + C$

PT $I = \int e^x \{f(x) + f'(x)\} dx = e^x f(x) + C$

$I = \int e^x f(x) dx + \int e^x f'(x) dx$

$I = I_1 + I_2$ ——— (1)

Now $I_1 = \int e^x f(x) dx$

$= f(x)(e^x) - \int \{f'(x).(e^x) dx$

$I_1 = e^x f(x) - I_2$

$I_1 + I_2 = e^x f(x)$

i.e. $I = e^x f(x) + C$

Q. $I = \int (x \log x) dx = ?$

So $I = \int (\log x) \cdot x \, dx$

$= \log x \left(\frac{x^2}{2}\right) - \int \left(\frac{1}{x}\right)\left(\frac{x^2}{2}\right) dx + C$

$= \frac{x^2}{2}(\log x) - \frac{1}{2}\left(\frac{x^2}{2}\right) + C$

Q. $I = \int (\log x) dx = ? = \int (\log x) \cdot x^0 dx$

$= \log x (x) - \int \left(\frac{1}{x}\right)(x) dx + C$

$= x \log x - x + C$

$$\boxed{\int \log x\,dx = x\log x - x}$$

(P8) $I = \int (\log_e x)^2\,dx = ?$

$$= \int (\log x)^2 \cdot 1\,dx$$

$$= (\log x)^2 (x) - \int \left(2\log x \times \frac{1}{x}\right)\cdot(x)\,dx + c$$

$$= x(\log x)^2 - 2\int (\log x)\,dx + c$$

$$= x(\log x)^2 - 2\left[x\log x - x\right] + c$$

$$\frac{d}{dx}(\log x)^2 = 2(\log x)^{2-1}\cdot\frac{1}{dx}(\log x)$$

$$= 2(\log x)\cdot\left(\frac{1}{x}\right)$$

Quest Based on (M-IV)

Q. $I=\int \frac{\cos\theta}{(2+\sin\theta)(3+4\sin\theta)}d\theta = ?$

Put $\sin\theta = t \Rightarrow \cos\theta\, d\theta = dt$

$$I=\int \frac{1}{(2+t)(3+4t)}dt$$

$$\frac{1}{(2+t)(3+4t)} = \frac{A}{2+t}+\frac{B}{3+4t}$$

$$1 = A(3+4t)+B(2+t)$$

Put $t=-2$, $1 = A(3-8) \Rightarrow A = -\frac{1}{5}$

Put $t=-\frac{3}{4}$, $1 = B\left(2-\frac{3}{4}\right) \Rightarrow B = \frac{4}{5}$

i.e $$\frac{1}{(2+t)(3+4t)} = \frac{-1/5}{2+t}+\frac{4/5}{3+4t}$$

$$= -\frac{1}{5}\int\frac{dt}{2+t}+\frac{4}{5}\int\frac{dt}{3+4t}$$

$$= -\frac{1}{5}\log(t+2)+\frac{1}{5}\int\frac{dt}{\left(t+\frac{3}{4}\right)} = -\frac{1}{5}\log(t+2)+\frac{1}{5}\log\left(t+\frac{3}{4}\right)+C$$

$$= -\frac{1}{5}\log(\sin\theta+2)+\frac{1}{5}\log\left(\sin\theta+\frac{3}{4}\right)+C$$

(PP) $I=\int \frac{x^2+x+1}{(x-1)^3}dx = ? = \int\left(\frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{(x-1)^3}\right)dx$ ≈ little bit lengthy.

(M-II) Put $x-1=t$, $dx=dt$ So $I=\int \frac{(t+1)^2+(t+1)+1}{t^3}dt = \int \frac{t^2+1+2t+t+2}{t^3}dt$

$$I=\int\left(\frac{t^2+3t+3}{t^3}\right)dt = \int\left(\frac{1}{t}+\frac{3}{t^2}+\frac{3}{t^3}\right)dt$$

$$= \log t + 3\left(\frac{-1}{t}\right)+3\left(\frac{-1}{2t^2}\right)+C$$

$$= \log(x-1)-\frac{3}{(x-1)}-\frac{3}{2(x-1)^2}+C$$

$$\int t^{-2}dt = \frac{t^{-2+1}}{-2+1} = \frac{t^{-1}}{-1} = \frac{-1}{t}$$

$$\int t^{-3}dt = \frac{t^{-3+1}}{-3+1} = \frac{t^{-2}}{-2} = \frac{-1}{2t^2}$$

HW8 $I = \int \frac{\sqrt{x}}{x+1}\,dx = ?$ ..... Ans: $2(\sqrt{x} - \tan^{-1}\sqrt{x}) + C$

HINT: Put $x = t^2$

$dx = 2t\,dt$


Thank
YOU